

Real and Virtual Leader-follower Strategies in Lane Changing, Merging and Overtaking Maneuvers

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Abstract—This paper designs a nonlinear, time-invariant motion planner that guarantees collision-free lane changing, merging and overtaking maneuvers, whilst considering all practical limitations and constraints due to the system nonholonomy, workspace geometry, and steering specifications. The motion planner derived from the Lyapunov-based Control Scheme works within an overarching leader follower scheme. We note that lane merging is accomplished through a host of strategies, *inter alia*, the robot from the merging lane is strategically assigned a separate target, and the platoon has its own target. The platoon overtaking is facilitated via virtual-leader technique, whereas single robot overtaking is facilitated via real-leader follower strategy. The effectiveness of the control laws is demonstrated via computer simulations. It also emphasizes the potential contribution to the intelligent vehicle system's discipline.

Keywords—Lyapunov-based Control Scheme; leader follower strategy; virtual-leader; nonholonomy; lane merging; lane changing maneuver

I. INTRODUCTION

The conditions of urban transportation have gone from bad to worse as the number of vehicles on our roads and highways increases on a daily basis. The problem is compounded by complicated vehicle maneuvers such as lane changing, lane merging and overtaking which are seen to be executed on a more frequent basis nowadays. A solution to this transportation problem is to develop an intelligent vehicle system (IVS) for our roads. This is a system of autonomous robots capable of complex and complicated maneuvers, without the provision of human envious in the equation. An intelligent vehicle is equipped with sensors, computers and a control system. The sensors enable the vehicle to perceive the road, potential hazards, and other vehicles; the computers process this information and determine actions (such as steering or braking); and the control system executes the chosen actions. The literature is inundated with functionalities and features added to the IVS in the quest to improve the autonomous system by improving safety, reducing congestion and

improving the driving conditions during, especially, the lane changing, merging and overtaking maneuvers [1, 2]. The work carried out in the lane changing and merging areas continues to find better and safer controllers and algorithms [2, 3, 4, 5, 6, 7]. In the interest of brevity, we mention only a few in this paper.

Rajamani, Tan, Law and Zhang [4] in 2000 used lane change to get an autonomous vehicle to automatically leave or join a platoon of unmanned vehicles circulating in a different lane. Swaroop and Yoon (1999) used lane change maneuver as a response to an emergency situation, treating as obstacle avoidance by presenting the design of a controller and a communication system to manage situation using the steering and speed controllers that were developed by Hedrick, McMahon, Narendran and Swaroop [8] in 1991. Naranjo, González, García and de Pedro [2] in 2008 carried out lane change by defining the estimated lane change trajectory and then using onboard sensors to track the path from input variables of a magnetic marker sequence. Ghumman, Kunwar and Benhabib [7] in 2008 presented a guidance based online trajectory planning algorithm for autonomous grounded vehicle overtaking maneuvers in dynamic highway environment focusing on time-optimal overtaking, obstacle avoidance and passenger comfort. The proposed algorithm used a modified Rendezvous Guidance method to obtain optimal vehicle acceleration commands for the overtaking maneuver. A shadow target concept was utilized to adjust in real time driving parameters of the obstacle vehicles in the driving lane as well as in the passing lane. Jula, Kosmatopoulos and Loannou [6] in 2000 and Wei, Dolan and Litkouhi [9] in 2010 presented a general algorithm to calculate the minimum longitudinal safety spacing for all vehicles involved. Sharma, Vanualailai and Chand [22] in 2008 considered motion planning of a flock of car-like robots which utilized the split/rejoin strategy to facilitate obstacle avoidance. Herein the leadership role was fixed to a single robot throughout the translation of the formation.

This paper focuses on lane changing, merging and overtaking maneuvers via a leader follower strategy. The leader follower strategy is perceived robust during lane changing and overtaking maneuvers since a collaborative

behavior is expected for a safe and collision free maneuverance. This strategy can be of two types: real and the virtual-leader follower strategies.

The real-leader follower approach encompasses a single robot splitting from the platoon formation in order to give way to the robot looming from the merging lane. Subsequently this robot overtakes the leader robot of the platoon, becoming the new leader of the platoon formation.

The entire platoon overtaking maneuver is accomplished through a virtual-leader follower strategy [10] and is represented by a point. The robot from the merging lane is strategically assigned a separate target to perform the lane merge maneuver, and the platoon has its own target.

In this paper, lane merge is considered as a driveway link to the main road. This paper also integrates the lane change and overtaking maneuvers into the path planning and control problem which has been addressed using different yet innovative strategies in literature [10, 11, 12, 13, 14, 15, 16, 17]. One of the recently designed artificial potential field methods, the Lyapunov-based Control Scheme (LbCS), will be utilized to achieve single robot and entire platoon lane changing and overtaking maneuvers.

II. BACKGROUND

The artificial potential field method designed by Khatib [18] in 1986 is commonly deployed in the area of robotics research and especially applied to the motion planning and control of various systems, including ones tagged with holonomic and/or nonholonomic constraints [10, 19, 20, 21]. This category of motion planners invariably boosts advantages such as easier analytical presentation of system singularities and inequalities, simplicity and elegance, favorable processing speed, decentralization and scalability features [16, 17]. A new control scheme using the potential fields was designed by Sharma [22] in 2008 known as the LbCS involves the design of attractive and repulsive potential field functions for attraction to the target and avoidance of various obstacles, respectively. The attractive functions are designed as distance measures which are considered as attractive potential field functions. We design each repulsive potential field function with positive tuning parameter in the numerator while the denominator contains an avoidance function, which is normally a distance measure between the centers of the mobile robot and obstacle. The entire workspace is then inundated with positive and negative fields, with the direction of motion facilitated via the notion of steepest decent [17]. For vehicular robotic systems, the gradient of the total artificial potential field, referred to as the input force, determines the speed and direction along which the robots move.

A. An Illustrative Example of the Lyapunov-based Control Scheme

This section presents a brief overview of LbCS utilized to generate artificial potential fields by considering the path planning problem of a point-mass reaching its target while avoiding a stationary object along its path.

Definition 1: The point-mass, PM is a disk of radius $r_{PM} \geq 0$ positioned at $(x(t), y(t)) \in \mathbf{R}^2$ at time $t \geq 0$. Precisely, the point-mass is the set

$$PM = \{(z_1, z_2) \in \mathbf{R}^2 : (z_1 - x)^2 + (z_2 - y)^2 \leq r_{PM}^2\},$$

with respect to the $z_1 z_2$ -plane.

1) Target Attraction

Definition 2: The target for the point-mass, PM is a disk with center (τ_1, τ_2) and radius $r_{TA} \geq 0$. It is the set

$$TA = \{(z_1, z_2) \in \mathbf{R}^2 : (z_1 - \tau_1)^2 + (z_2 - \tau_2)^2 \leq r_{TA}^2\}.$$

For the point-mass to be attracted to its target configuration, an attractive potential field function is designed. This is a measure of the Euclidean distance between the point-mass PM and its target TA .

$$H(x, y) = \frac{1}{2} [(x - \tau_1)^2 + (y - \tau_2)^2].$$

2) Obstacles Avoidance

Definition 3: The stationary solid object, SO is a circular disk with centre (o_1, o_2) and radius $r_{SO} \geq 0$. Precisely, the solid object is defined as the set

$$SO = \{(z_1, z_2) \in \mathbf{R}^2 : (z_1 - o_1)^2 + (z_2 - o_2)^2 \leq r_{SO}^2\}.$$

The solid object is treated as a fixed obstacle and an appropriate obstacle avoidance function is designed for its avoidance. This is a measure of the Euclidean distance between the centers of the point-mass PM and the fixed obstacle, SO and is given as

$$F_O(x, y) = \frac{1}{2} [(x - o_1)^2 + (y - o_2)^2 - (r_{PM} + r_{SO})^2].$$

Given some positive real value of the parameter $\alpha > 0$ a corresponding repulsive potential field function

$$\frac{\alpha}{F_O(x, y)}$$

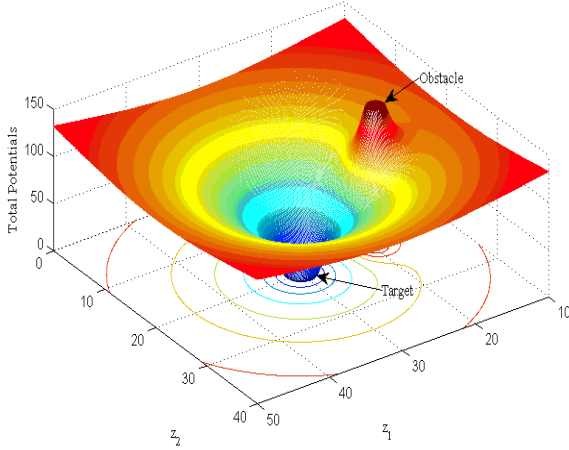
is constructed in accordance to the LbCS.

3) Total Potentials

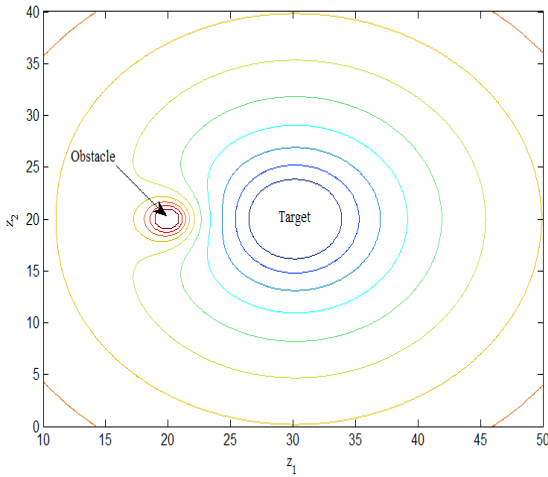
The total potentials is the sum of the attractive and repulsive potential field functions given as

$$L(x, y) = H(x, y) + \frac{\alpha}{F_o(x, y)}. \quad (1)$$

Fig. 1 shows the total potentials generated by Eq. (1) for $\alpha = 0.5$, with the target position at $(30, 20)$ and an obstacle fixed at $(20, 20)$. Fig. 1(a) shows a 3D visualization while its contour plot is shown in Fig. 1(b).



(a) 3D Visualization



(b) Contour Plot

Fig. 1 shows the total potentials and the contour plot generated by Eq. (1).

III. PLATOON MODEL

Definition 4: A platoon consists of a convoy of n virtually connected nonholonomic car-like mobile robots. An i th robot in the platoon is a circular disk with radius r_{vi} and positioned at center (x_i, y_i) . The i th car-like robot is the set

$A_i = \{(z_1, z_2) \in \mathbf{R}^2 : (z_1 - x_i)^2 + (z_2 - y_i)^2 \leq r_{vi}^2\}$ for $i \in \{1, \dots, n\}$, where A_i embodies a rear-wheel driven and front-wheel steered vehicle. The schematic diagram of A_i is seen in Fig. 2. We let m_i be the mass of the robot, F_i the force along the axis of the robot, Γ_i the torque about a vertical axis at $[x_i, y_i]^T$ and I_i the moment of inertia of the robot, then the dynamic model of A_i with respect to its CoM is

$$\left. \begin{aligned} \dot{x}_i &= v_i \cos \theta_i - \frac{l_1}{2} \omega_i \sin \theta_i, & \dot{y}_i &= v_i \sin \theta_i + \frac{l_1}{2} \omega_i \cos \theta_i, \\ \dot{\theta}_i &= \frac{v_i}{l_1} \tan \theta_i := \omega_i, & \dot{v}_i &= \frac{F_i}{m_i} := \sigma_i, & \dot{\omega}_i &= \frac{\Gamma_i}{I_i} := \eta_i, \end{aligned} \right\} \quad (2)$$

v_i and ω_i are the instantaneous translational and rotational velocities of A_i , while its instantaneous translational and rotational accelerations, the nonlinear acceleration-based control laws, are given as σ_i and η_i , respectively.

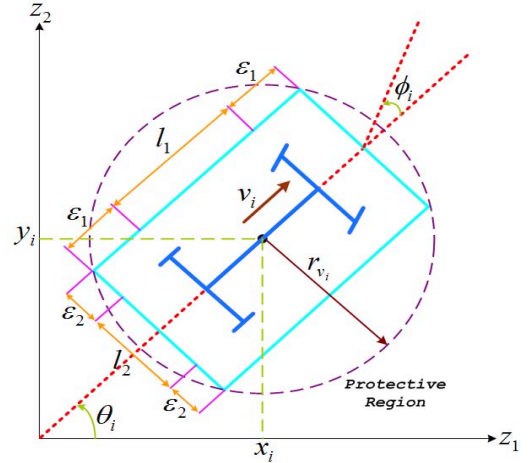


Fig. 2. Kinematic model of the car-like mobile robot A_i (re-constructed from Sharma, Vanualailai and Chand [23]).

We note that system (2) appropriately reflects the nonholonomy of the robots. With reference to Fig. 2, $[x_i, y_i]^T$ denotes the CoM of A_i , ϕ_i gives its steering wheel's angle with respect to the longitudinal axis, l_1 is the distance between the centre of the rear and front axles, while l_2 is the length of each axle. The configuration of A_i is given by $\mathbf{q}_i = [x_i, y_i, \theta_i]^T \in \mathbf{R}^3$, while $\mathbf{d}_i = [x_i, y_i]^T \in \mathbf{R}^2$ is its position and $\theta_i \in \mathbf{R}$ its angle of orientation with respect to z_1 -axis.

The state of robot A_i is captured in

$\mathbf{x}_i = [x_i, y_i, \theta_i, v_i, \omega_i]^T \in \mathbf{R}^5$ and its acceleration controls

represented by $\mathbf{u}_i = [\sigma_i, \eta_i]^T \in \mathbf{R}^2$. The states of all the n robots and the acceleration controls are collected in the vector forms $\mathbf{x} = [\mathbf{x}_1^T, \dots, \mathbf{x}_n^T] \in \mathbf{R}^{5 \times n}$ and $\mathbf{u} = [\mathbf{u}_1^T, \dots, \mathbf{u}_n^T]^T \in \mathbf{R}^{2 \times n}$, resp.

Given the *clearance parameters* ϵ_1 and ϵ_2 each A_i is enclosed in a protective circular region centered at $\mathbf{d}_i = [x_i, y_i]^T \in \mathbf{R}^2$, radius $r_i = \sqrt{(l_1 + 2\delta_1)^2 + (l_2 + 2\delta_2)^2} / 2$. This representation not only maximizes the free space available to the robots in the workspace but also ensures easier construction of the potential field functions [19, 23].

IV. LEADER FOLLOWER STRATEGY

A new leader follower scheme is designed by modifying the one constructed by Sharma, Vanualailai and Prasad [19] and utilized to maintain the prescribed formation and execute lane changing and overtaking maneuvers. Fig. 3 illustrates the overview of new leader follower strategy and introduces new measures $L_i \cos \psi_i$, $L_i \sin \psi_i > 0$, for $i = 1$ to n . The *mobile ghost targets*, are positioned relative to the leader robot with a user-defined Euclidean measure of $L_i \cos \psi_i$ units right or left and $L_i \sin \psi_i$ units up or down, while the centre of the ghost target is given by $(x_1 + L_i \cos \psi_i, y_1 + L_i \sin \psi_i)$. This modified scheme includes an angular measure between the leader and the follower robots which now allows for preferred orientation of the moving ghost target. In this pre-determined prescribed formation, the leader robot moves towards its target (τ_{i1}, τ_{i2}) while the ghost targets move relative to the Euclidean position of the leader robot, and in turn the follower robots are attracted towards their predefined targets (mobile ghost target) for all $t > 0$. It also helps to maintain the prescribed formation. The newly defined parameters L_i and ψ_i are defined in piecemeal for specific stages.

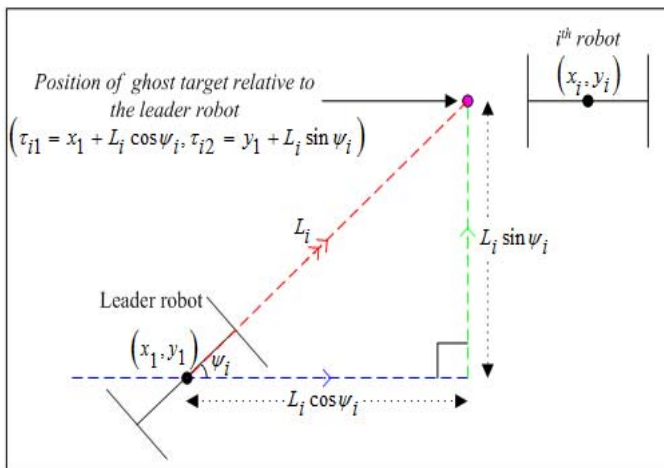


Fig. 3 A schematic diagram of a ghost target - leader robot setup, (modified from Sharma, Vanualailai and Prasad [19]).

Using the new parameters, the following bounding functions between the position coordinates of the leader robot and the i th follower robot are designed:

$$\tau_{i1} = x_1 + L_i \cos \psi_i, \quad \tau_{i2} = y_1 + L_i \sin \psi_i. \quad (3)$$

V. FORMULATION OF THE PROBLEM

The section begins by making the following assumptions and definitions:

Definition 5: A real-leader is a visible member robot of the platoon, which contributes to the movement of the convoy via the mobile ghost targets tagged to it.

Definition 6: A virtual-leader is represented as a point which is an invisible leader of a platoon. Through the tagged ghost targets defined in piecemeal stages, it accomplish lane changing (give-way) and overtaking maneuvers.

Assumption 1: Each car-like mobile robot will be able to detect the other car-like mobile robots in its vicinity using sensors. The integrated sensors also detect the road lanes and boundaries of road lanes.

Definition 7: Lane merge is a driveway link to the main road facilitating a lane change.

Remark 1: Lane change maneuver is a result of:

- a lane merge or;
- a robot/robots in a platoon formation giving way to the robot looming from the merging lane or;
- overtaking.

Remark 2: An overtaking maneuver is the outcome of a lane change where a mobile robot accelerates to overtake a robot in the other lane. This robot may later return safely to its original lane.

This new leader follower strategy maintains the prescribed formation and facilitates the maneuvers. A normal highway includes multiple lanes as well as occasional feeder lanes. These lanes are separated by white-painted distinct segments, which guide the vehicles in their lane while the boundaries of road lanes are called edge lines. The edge lines outline and separate the outside edge of the pavement from the road lanes. In LbCS, these linear segments are considered as fixed obstacles which need to be avoided in order to restrict robots in specific lanes. Then there are moving and artificial obstacles. The moving obstacles are the robots themselves and the artificial obstacles are the dynamic constraints of the robotic system.

The research problem is illustrated with two scenarios:

A. Scenario 1: Single robot giving way and overtaking via a real-leader

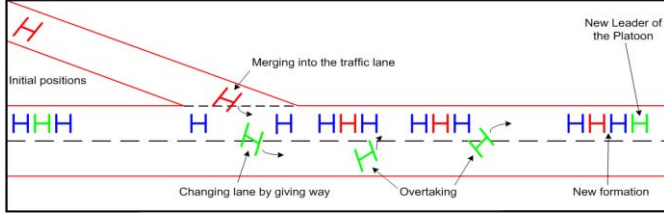


Fig. 4 Schematic representation of a single robot (represented in color green) giving way to the robot (red) from the merging lane and overtaking via a real-leader.

As shown in Fig. 4, a single pre-selected robot from the platoon in the adjacent lane splits to accommodate the robot approaching from the merging lane. The pre-selected robot gives way to the robot from the merging lane. Along the course of the journey, this robot overtakes the leader robot of the platoon, aligns to a new platoon formation becoming the new leader of the platoon. We note that any robot from the platoon can potentially give way to the robot looming from the merging lane. The appropriate choice of the ghost targets and the tuning parameters allow this selection. In the scenario a lane merge maneuver is advocated through two different strategies: (1) establishing a separate target position for the robot looming from the merging lane and (2) establishing a separate target for the leader robot.

B. Scenario 2: Platoon overtaking via a virtual-leader

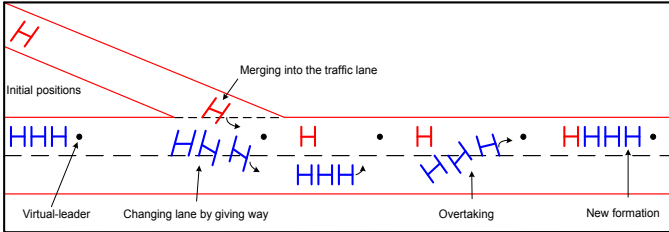


Fig. 5. Schematic diagram of the entire platoon giving way and then overtaking via virtual-leader which is represented by a ghost target.

Fig. 5 shows a scenario where an entire platoon in the adjacent lane accommodates a robot looming from the merging lane by changing its original lane. Finally, the platoon overtakes the robot from the merging lane and aligns to form a new formation. The give-way and overtaking maneuvers are accomplished via the virtual-leader follower strategy. The virtual-leader is represented by a point and acts as a ghost target for the platoon. The virtual-leader has its own target in this new strategy.

VI. DESIGN OF POTENTIAL FIELD FUNCTIONS

Our control objective is to design acceleration based controllers for each car in system (2), within the framework of LbCS. The car-like mobile robots are required to carry out lane change, merge and overtaking maneuvers with obstacle and collision avoidances.

A. Establish Prescribed Formation

Specific selection of the initial positions and orientations of the car-like robotic system establish the desired platoon formation, which includes the order of cars within the formation.

B. Attraction to Target and Ghost Targets

To initiate movement of a robot from the initial position in the merging lane and the platoon of robots in the adjacent lane, a target (or ghost target) is proposed for each robot A_i :

$$T_i = \left\{ (x_i, y_i) \in \mathbf{R}^2 : (x_i - \tau_{i1})^2 + (y_i - \tau_{i2})^2 \leq r_{t_i}^2 \right\} \text{ with center}$$

(τ_{i1}, τ_{i2}) and radius r_{t_i} (adopted from Sharma, Vanualailai and Chand [23]). For the car-like robots to be attracted to their respective targets, an attractive potential field function is designed as $U_{att_i} : \mathbf{R}^4 \rightarrow \mathbf{R}^+$ with

$$U_{att_i}(\mathbf{x}) = \sum_{i=1}^n \frac{1}{2} \ln(H_i(\mathbf{x}) + 1), \quad (4)$$

where the associated target attractive function is of the form

$$H_i(\mathbf{x}) = \frac{1}{2} \left\{ (x_i - \tau_{i1})^2 + (y_i - \tau_{i2})^2 + v_i^2 + \omega_i^2 \right\}, \text{ for } i \in \{1, \dots, n\}, \quad n \in \mathbf{N}. \quad (5)$$

The role of the function would be to establish positive potential fields around the i th target and to induce maneuvers of the designated car-like robots towards it. The leader robot and the robot looming from the merging lane are assigned individual targets while the follower robots have ghost targets.

C. Path Design

The path design comprises of a piecewise integration of linear segments, which are treated as fixed obstacles.

1) Fixed Obstacles: Markings of road lanes and associated boundaries

Definition 10: The lane markings and boundaries of road lanes, either real or virtual are collapsed into straight line segments in the $z_1 z_2$ -plane. The parametric representation of the k th segment with initial coordinates (q_{k1}, r_{k1}) and final

coordinates (q_{k2}, r_{k2}) can be given as $x_k = q_{k1} + \lambda_k (q_{k2} - q_{k1})$ and $y_k = r_{k1} + \lambda_k (r_{k2} - r_{k1})$ where $\lambda_k : \mathbf{R} \rightarrow [0, 1]$.

As shown in Fig. 6, the problem of collision avoidance for the entire maneuverance is solved by treating each segment as a fixed obstacle.

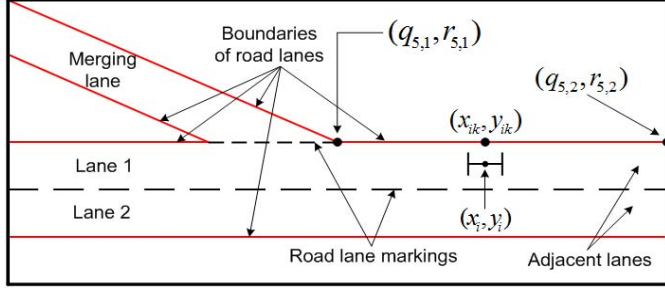


Fig. 6 A schematic diagram of the piecewise integration of road lanes. The center (x_i, y_i) of every A_i , for $i \in \{1, \dots, n\}$ will avoid the (x_{ik}, y_{ik}) point on the k th linear segment.

We adopt the minimum distance technique (MDT) from Sharma, Vanualailai and Singh [17] to facilitate the avoidance of these straight line segments. MDT involves calculation of the minimum distance from the mobile robot to the line segment and the avoidance of the closest point. Avoidance of the closest point on the line at any time $t \geq 0$ essentially results in the avoidance of the entire line segment. From geometry, coordinates of the closest point can be given as $x_{ik} = q_{k1} + \lambda_{ik} (q_{k2} - q_{k1})$ and $y_{ik} = r_{k1} + \lambda_{ik} (r_{k2} - r_{k1})$, where $\lambda_{ik} = (x_i - q_{k1})c_k + (y_i - r_{k1})d_k$,

$$c_k = \frac{(q_{k2} - q_{k1})}{(q_{k2} - q_{k1})^2 + (r_{k2} - r_{k1})^2}, \quad d_k = \frac{(r_{k2} - r_{k1})}{(q_{k2} - q_{k1})^2 + (r_{k2} - r_{k1})^2},$$

and the saturation function $\lambda_{ik} \in \mathbf{R}$ is

$$\lambda_{ik}(x_i, y_i) = \begin{cases} 0, & \text{if } \lambda_{ik} < 0 \\ \lambda_{ik}, & \text{if } 0 \leq \lambda_{ik} \leq 1 \\ 1, & \text{if } \lambda_{ik} > 1. \end{cases}$$

Note that $\lambda_{ik}(x_i, y_i)$ is a nonnegative scalar such that it is restricted to the interval $[0, 1]$. Hence, there is always an avoidance of the k th segment of the road lanes at every iteration $t \geq 0$. The reader is referred to Sharma, Vanualailai and Singh [17] for a detailed explanation of MDT. For avoidance of such points by A_i the tuning parameter $\alpha_{ik} > 0$ is introduced and the repulsive potential field function defined by $U_{rep_{i1}} : \mathbf{R}^2 \rightarrow \mathbf{R}^+$ with

$$U_{rep_{i1}}(\mathbf{x}) = \sum_{k=1}^p \frac{\alpha_{ik}}{L_{ik}(\mathbf{x})}, \quad (6)$$

where the obstacle avoidance function is $\lambda_{ik} \in \mathbf{R}$ is

$$L_{ik}(\mathbf{x}) = \frac{1}{2} \left[(x_i - x_{ik})^2 + (y_i - y_{ik})^2 - r_v^2 \right], \quad \text{for} \\ i \in \{1, \dots, n\}, \quad k \in \{1, \dots, p\}, \quad n, p \in \mathbf{N}. \quad (7)$$

Figure 7 shows the potential fields and the corresponding contour plot generated from the attractive function governed by eqn. (4) and the repulsive potential field function governed by eqn. (7). This is achieved by the strategic selection of the numerical values of the individual linear segments comprising into road lanes where the target is placed at $(60, 5)$.

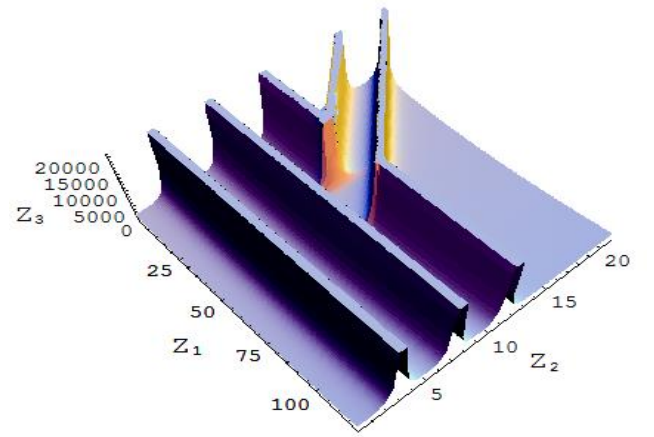


Fig. 7 (a). 3D visualization of the Total Potentials

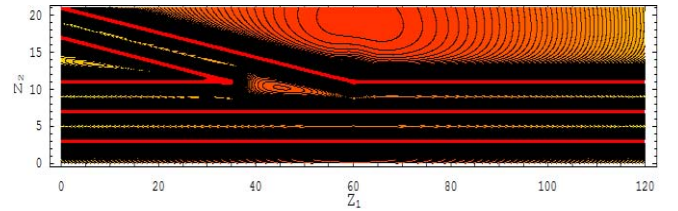


Fig. 7 (b) Contour Plot

VII. INTER-ROBOT AVOIDANCE

A car itself becomes a moving obstacle for all other car-like robots in the workspace. Hence, the integrated sensors of each robot detect the other robots and a two-way avoidance scheme is deployed. This allows collision avoidance of every A_i and A_j robots while in the platoon, and during the changing lanes and overtaking maneuvers.

To prevent inter-robot collisions the bound $\|d_i - d_j\|^2 > N_{ij}^2$ is included, where N_{ij} is the minimum

Euclidean distance between every A_i and A_j , and given as $(r_{v_i} + r_{v_j})^2 = 4r_v^2$ on $\mathbf{R}^2$, assuming $r_{v_i} = r_{v_j}$. This bound is treated in the LbCS by developing an appropriate artificial obstacle, $AO_{ij} = \{\mathbf{x} \in \mathbf{R}^2 : (x_i - x_j)^2 + (y_i - y_j)^2 \leq N_{ij}^2 = 4r_v^2\}$.

For avoidance of these artificial obstacles by each A_i , a tuning parameter $\gamma_{ij} > 0$ is introduced, and a repulsive potential field function defined by $U_{rep_{i2}} : \mathbf{R}^2 \rightarrow \mathbf{R}^+$ with

$$U_{rep_{i2}}(\mathbf{x}) = \sum_{\substack{j=0 \\ j \neq i}}^n \frac{\gamma_{ij}}{V_{ij}(\mathbf{x})}, \quad (8)$$

where the obstacle avoidance function is of the form

$$V_{ij}(\mathbf{x}) = \frac{1}{2} \left[(x_i - x_j)^2 + (y_i - y_j)^2 - 4r_v^2 \right], \text{ for } j \in \{1, \dots, n\} \setminus \{i\}, \quad n \in \mathbf{N}. \quad (9)$$

VIII. DYNAMIC CONSTRAINTS

The constraints imposed on the translational, v_i and rotational, ω_i velocities [17, 23] are:

- $|v_i| \leq v_{\max}$, where $v_{\max} > 0$ is the maximum achievable speed of the mobile robot;
- $|\omega_i| \leq \frac{|v_i|}{\rho_{\min}} \leq \frac{v_{\max}}{\rho_{\min}}$, is derived from $v_i^2 \geq \rho_{\min}^2 \omega_i^2$ where $\rho_{\min}$ is known as the *minimum turning radius*

and is given as $\rho_{\min} = \frac{l_i}{\tan \phi_{\max}}$.

To ensure that the car-like mobile robots operate within the dynamic constraints the following *artificial obstacles* are adopted from Sharma, Vanualailai and Chand [23]:

$$DO_{i1} = \{v_i \in \mathbf{R} : v_i \leq -v_{\max} \text{ or } v_i \geq v_{\max}\},$$

$$DO_{i2} = \{\omega_i \in \mathbf{R} : \omega_i \leq -v_{\max} / |\rho_{\min}| \text{ or } \omega_i \geq v_{\max} / |\rho_{\min}|\}.$$

For avoidance of these artificial obstacles a tuning parameter $\beta_{is} > 0$ is introduced. The repulsive potential field function is given by $U_{rep_{i3}} : \mathbf{R}^2 \rightarrow \mathbf{R}^+$ with

$$U_{rep_{i3}}(\mathbf{x}) = \sum_{s=1}^2 \frac{\beta_{is}}{U_{is}(\mathbf{x})}, \quad (10)$$

where the associated avoidance functions (adopted from Sharma, Vanualailai and Chand [23]) are:

$$U_{i1}(\mathbf{x}) = \frac{1}{2} (v_{\max} - v_i)(v_{\max} + v_i), \quad (11)$$

$$U_{i2}(\mathbf{x}) = \frac{1}{2} \left(\frac{v_{\max}}{|\rho_{\min}|} - \omega_i \right) \left(\frac{v_{\max}}{|\rho_{\min}|} + \omega_i \right), \text{ for } i \in \{1, \dots, n\}, \quad n \in \mathbf{N}. \quad (12)$$

IX. CONTROLLER DESIGN

A. Auxiliary Function

To ensure that the total potentials vanish when the robots converge to their final target configurations, an auxiliary function of the form

$$G_i(\mathbf{x}) = \frac{1}{2} \left\{ (x_i - \tau_{i1})^2 + (y_i - \tau_{i2})^2 + (\theta_i - \tau_{i3})^2 \right\}, \quad (13)$$

is introduced, where τ_{bi3} is the desired final orientation of A_i .

B. Designing of Controllers

The summing of all attractive and repulsive potential field functions yield the following Lyapunov function or total potential for system (2):

$$L_{(1)}(\mathbf{x}) = \sum_{i=0}^n \left[H_i(\mathbf{x}) + G_i(\mathbf{x}) \left(\sum_{k=1}^p \frac{\alpha_{ik}}{L_{ik}(\mathbf{x})} + \sum_{s=1}^2 \frac{\beta_{is}}{U_{is}(\mathbf{x})} + \sum_{\substack{j=1 \\ j \neq i}}^n \frac{\gamma_{ij}}{V_{ij}(\mathbf{x})} \right) \right]. \quad (14)$$

The time-derivative of L along every solution of system (2) is the dot product of the gradient of L , given by

$$\nabla L = \left(\frac{\partial L}{\partial x_1}, \frac{\partial L}{\partial y_1}, \frac{\partial L}{\partial \theta_1}, \frac{\partial L}{\partial v_1}, \frac{\partial L}{\partial \omega_1}, \dots, \frac{\partial L}{\partial x_n}, \frac{\partial L}{\partial y_n}, \frac{\partial L}{\partial \theta_n}, \frac{\partial L}{\partial v_n}, \frac{\partial L}{\partial \omega_n} \right),$$

and the time-derivative of the vector $\mathbf{x} = (x_1, y_1, \theta_1, v_1, \omega_1, \dots, x_n, y_n, \theta_n, v_n, \omega_n)$. Finding the dot product $(\nabla L_{(1)} \times \dot{\mathbf{x}} := \dot{L}_{(1)}(\mathbf{x}))$ along a particular trajectory of system (2), and upon collecting terms with v_i and ω_i separately, give:

$$\begin{aligned} \dot{L}_{(1)}(\mathbf{x}) = & \sum_{i=1}^n [(f_{1i} \cos \theta_i + f_{2i} \sin \theta_i + f_{4i} \sigma_i) v_i \\ & + \left(\frac{l_2}{2} (f_{2i} \cos \theta_i - f_{1i} \sin \theta_i) + f_{3i} + f_{5i} \eta_i \right) \omega_i] \end{aligned} \quad (15)$$

where the functions f_{1i} to f_{5i} , for $i=1, \dots, n$, are defined as (on suppressing the variable $\mathbf{x}$):

$$f_{3i} = \left(\sum_{k=1}^p \frac{\alpha_{ik}}{L_{ik}} + \sum_{s=1}^2 \frac{\beta_{is}}{U_{is}} + \sum_{\substack{j=1 \\ j \neq i}}^n \left(\frac{\gamma_{ij}}{V_{ij}} + \frac{\gamma_{ji}}{V_{ji}} \right) \right) (\theta_i - \tau_{i3}),$$

$$f_{4i} = \frac{1}{H_i + 1} + G_i \frac{\beta_{i1}}{U_{i1}^2}, \quad f_{5i} = \frac{1}{H_i + 1} + G_i \frac{\beta_{i2}}{U_{i2}^2}.$$

For $i=2, \dots, n$,

$$\begin{aligned} f_{1i} = & \left[\left(\frac{1}{H_i + 1} + \sum_{k=1}^p \frac{\alpha_{ik}}{L_{ik}} + \sum_{s=1}^2 \frac{\beta_{is}}{U_{is}} + \sum_{\substack{j=1 \\ j \neq i}}^n \left(\frac{\gamma_{ij}}{V_{ij}} + \frac{\gamma_{ji}}{V_{ji}} \right) \right) (x_i - \tau_{i1}) \right. \\ & - G_i \sum_{k=1}^p \frac{\alpha_{ik}}{L_{ik}^2} [(x_i - (q_{k1} + \lambda_{ik}(q_{k2} - q_{k1}))) (1 - c_k(q_{k2} - q_{k1}))] \\ & - [c_k(y_i - (r_{k1} + \lambda_{ik}(r_{k2} - r_{k1}))) (r_{k2} - r_{k1})] \\ & \left. - G_i \sum_{\substack{j=1 \\ j \neq i}}^n \frac{\gamma_{ij}}{V_{ij}^2} (x_i - x_j) + G_j \sum_{\substack{j=1 \\ j \neq i}}^n \frac{\gamma_{ji}}{V_{ji}^2} (x_j - x_i), \right. \\ f_{2i} = & \left[\frac{1}{H_i + 1} + \sum_{k=1}^p \frac{\alpha_{ik}}{L_{ik}} + \sum_{s=1}^2 \frac{\beta_{is}}{U_{is}} + \sum_{\substack{j=1 \\ j \neq i}}^n \left(\frac{\gamma_{ij}}{V_{ij}} + \frac{\gamma_{ji}}{V_{ji}} \right) \right] (y_i - \tau_{i2}) \\ & - G_i \sum_{k=1}^p \frac{\alpha_{ik}}{L_{ik}^2} [(y_i - (r_{k1} + \lambda_{ik}(r_{k2} - r_{k1}))) (1 - d_k(r_{k2} - r_{k1}))] \\ & - [d_k(x_i - (q_{k1} + \lambda_{ik}(q_{k2} - q_{k1}))) (q_{k2} - q_{k1})] \\ & \left. - G_i \sum_{\substack{j=1 \\ j \neq i}}^n \frac{\gamma_{ij}}{V_{ij}^2} (y_i - y_j) + G_j \sum_{\substack{j=1 \\ j \neq i}}^n \frac{\gamma_{ji}}{V_{ji}^2} (y_j - y_i), \right] \end{aligned}$$

Moreover,

$$\begin{aligned} f_{11} = & \left[\left(\frac{1}{H_1 + 1} + \sum_{k=1}^p \frac{\alpha_{1k}}{L_{1k}} + \sum_{s=1}^2 \frac{\beta_{1s}}{U_{1s}} + \sum_{\substack{j=1 \\ j \neq 1}}^n \left(\frac{\gamma_{1j}}{V_{1j}} + \frac{\gamma_{j1}}{V_{j1}} \right) \right) (x_1 - \tau_{11}) \right. \\ & - G_1 \sum_{k=1}^p \frac{\alpha_{1k}}{L_{1k}^2} [(x_1 - (q_{k1} + \lambda_{1k}(q_{k2} - q_{k1}))) (1 - c_k(q_{k2} - q_{k1}))] \\ & - [c_k(y_1 - (r_{k1} + \lambda_{1k}(r_{k2} - r_{k1}))) (r_{k2} - r_{k1})] \\ & \left. - G_1 \sum_{\substack{j=1 \\ j \neq 1}}^n \frac{\gamma_{1j}}{V_{1j}^2} (x_1 - x_j) + G_j \sum_{\substack{j=1 \\ j \neq 1}}^n \frac{\gamma_{j1}}{V_{j1}^2} (x_j - x_1), \right. \\ f_{21} = & \left[\frac{1}{H_1 + 1} + \sum_{k=1}^p \frac{\alpha_{1k}}{L_{1k}} + \sum_{s=1}^2 \frac{\beta_{1s}}{U_{1s}} + \sum_{\substack{j=1 \\ j \neq 1}}^n \left(\frac{\gamma_{1j}}{V_{1j}} + \frac{\gamma_{j1}}{V_{j1}} \right) \right] (y_1 - \tau_{12}) \\ & - G_1 \sum_{k=1}^p \frac{\alpha_{1k}}{L_{1k}^2} [(y_1 - (r_{k1} + \lambda_{1k}(r_{k2} - r_{k1}))) (1 - d_k(r_{k2} - r_{k1}))] \\ & - [d_k(x_1 - (q_{k1} + \lambda_{1k}(q_{k2} - q_{k1}))) (q_{k2} - q_{k1})] \\ & \left. - G_1 \sum_{\substack{j=1 \\ j \neq 1}}^n \frac{\gamma_{1j}}{V_{1j}^2} (y_1 - y_j) + G_j \sum_{\substack{j=1 \\ j \neq 1}}^n \frac{\gamma_{j1}}{V_{j1}^2} (y_j - y_1), \right] \end{aligned}$$

We let the following theorem to introduce the control laws of system (2):

Theorem 1: Consider a robot in the merging lane and a platoon of nonholonomic car-like robots in the adjacent lane, where the motion is governed by ODEs described by system (2). The objective is to facilitate lane changing and overtaking maneuvers via real and virtual-leader follower strategies arising from lane merge maneuvers. Utilizing the potential field functions and introducing real numbers $\delta_{i1}, \delta_{i2} > 0$ commonly known as convergence parameters the following nonlinear continuous acceleration control laws can be generated for $A_i, i=1, \dots, n$:

$$\left. \begin{aligned} \sigma_i = & \frac{-(\delta_{i1} v_i + f_{1i}(\mathbf{x}) \cos \theta_i + f_{2i}(\mathbf{x}) \sin \theta_i)}{f_{4i}(\mathbf{x})}, \\ \eta_i = & \frac{-(\delta_{i2} \omega_i + \frac{l_1}{2} f_{2i}(\mathbf{x}) \cos \theta_i + f_{3i}(\mathbf{x}) - \frac{l_1}{2} f_{1i}(\mathbf{x}) \sin \theta_i)}{f_{5i}(\mathbf{x})}. \end{aligned} \right\} \quad (16)$$

X. PROOF OF SYSTEM STABILITY

Let $f: D \subset \mathbf{R}^{5 \times n}$ where $\mathbf{R}^{5 \times n}$ is assumed to be smooth enough to guarantee existence, uniqueness and continuous dependence of solutions of the system (2) in D , an open set in $\mathbf{R}^{5 \times n}$. For the purpose of stability analysis of the system (2), the following is considered:

Assumption 2: Let

$\mathbf{x}^* = (\tau_{11}, \tau_{12}, \tau_{13}, 0, 0, \dots, \tau_{n1}, \tau_{n2}, \tau_{n3}, 0, 0) \in \mathbf{R}^{5 \times n} \in D(L_{(2)})(\mathbf{x}) := D$ be an equilibrium point of system (2).

Remark 3: If $\dot{L}_{(2)}(\mathbf{x}^*) = f(\mathbf{x}^*) = 0$ making $\mathbf{x}^*$ a feasible equilibrium point, at least, in a small neighborhood of the target configuration.

The stability of the system can be easily verified:

1) $\dot{L}_{(2)}(\mathbf{x})$ is continuous and positive on the domain D given as

$$D(L_{(2)}(\mathbf{x})) = \{\mathbf{x} \in \mathbf{R}^{5 \times n} : L_{ik}(\mathbf{x}) > 0, k = 1, \dots, p, U_{is}(\mathbf{x}) > 0, s = 1, 2, \text{ and } V_{ij}(\mathbf{x}) > 0, j \neq i, j = 1, \dots, n, \text{ for } i = 1, \dots, n\}.$$

2) $L_{(2)}(\mathbf{x}^*) = 0, \mathbf{x}^* \in D$.

3) $L_{(2)}(\mathbf{x}) > 0, \forall \mathbf{x} \in D, \mathbf{x} \neq \mathbf{x}^*$.

$$4) \dot{L}_{(2)}(\mathbf{x}) = -\sum_{i=1}^n (\delta_{i1} v_i^2 + \delta_{i2} \omega_i^2) \leq 0, \forall \mathbf{x} \in D.$$

Finally, it can easily be shown that the first partials of $L_{(2)}(\mathbf{x})$ is C^1 which makes up the fifth condition of a Lyapunov function. Hence, the following theorem concludes the discussion on system stability.

Theorem 2: The equilibrium point $\mathbf{x}^*$ of system (2) is at least stable in the sense of Lyapunov provided the controllers σ_i and $\eta_i, i = 1, \dots, n$, are defined as in (16).

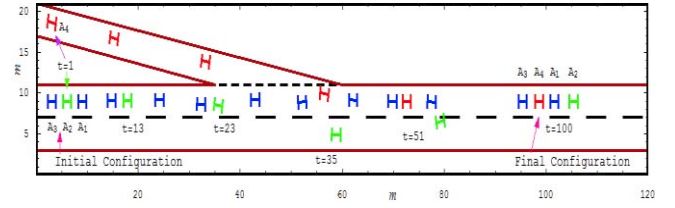
Remark 4: System (2) is guaranteed stability and not the asymptotic stability. This is in line with the celebrated Brockett's Theorem. Therefore, there is a possibility that the vehicle can stop in a small neighborhood of its goal. The stabilization of nonholonomic mechanical systems is also not possible with continuous controls, merely stability in practical situations suffice.

XI. IMPLEMENTATION OF CONTROL LAWS

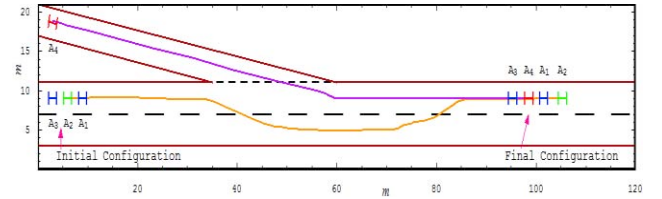
This section demonstrates the simulation results of lane changing and overtaking maneuvers arising from lane merge of car-like mobile robots guided by leader follower strategies. The overtaking maneuver of a robot splitting from the platoon formation in the adjacent lane is acquired by a real-leader follower strategy, while entire platoon overtaking maneuver is accomplished via a virtual-leader follower strategy. The lane merge is attained by assigning a separate target configuration for the robot looming from the merging lane. The scenarios

capture real traffic situations to show the effectiveness of the control laws proposed in this paper.

A. Scenario 1: Single Robot Giving Way and Overtaking via Real-Leader follower Scheme



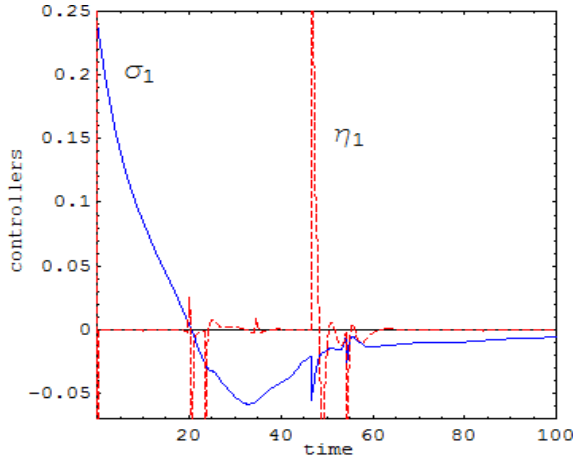
(a) A robot splits from the platoon formation to give way to the robot approaching from the merging lane, and overtakes the leader robot (A_1). Robot (A_4) from merging lane – represented by red, robot giving way (A_2) – represented by green, follower robot (A_3) – represented by blue.



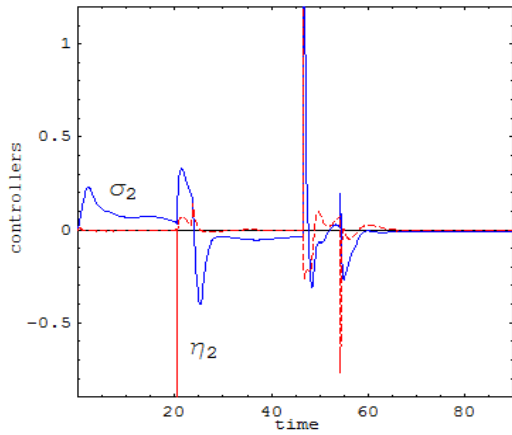
(b) The trajectory path generated by the single robot giving way and overtaking scenario. Path of the robot (A_4) from merging lane – represented by purple and the path of the robot giving way and overtaking (A_2) – represented by orange.

Fig. 8 shows the simulation result of single robot overtaking.

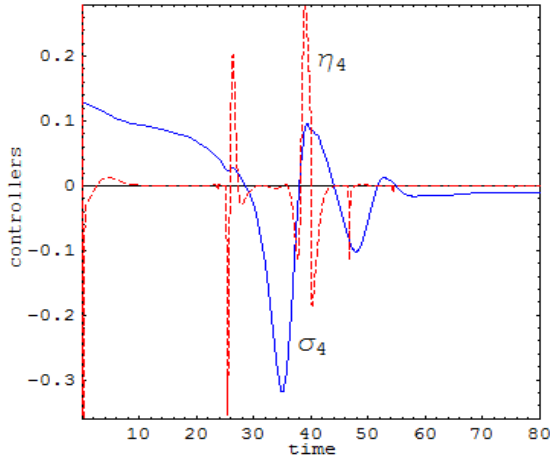
In this scenario, we show a nominated robot peeling from the platoon in the adjacent lane by giving way to the robot approaching from the merging lane. The former subsequently overtakes the leader robot, aligns in its original lane forming a new final platoon formation and becomes the new leader of the platoon (see Fig. 8 (a)). The maneuver is achieved via attraction forces of new leader follower strategy and the pre-defined final configurations. The path generated by the respective robots in this scenario is shown in Fig. 8 (b).



(a) Evolution of the acceleration components of A_1
(η_1 - translational, σ_1 - rotational).



(b) Evolution of the acceleration components of A_2
(η_2 - translational, σ_2 - rotational).



(c) Evolution of the acceleration components of A_4
(η_4 - translational, σ_4 - rotational).

Fig. 9 shows the evolution of the acceleration components of the leader

robot, robot peeling from the platoon and the robot looming from the merging lane.

The strategic selection of the numerical values of the ghost targets contributes to the lane changing and overtaking maneuvers. Fig. 9(a), 9(b) and 9(c) shows the evolution of acceleration components of A_1 - leader robot, A_2 - robot peeling from the platoon, and A_4 - robot looming from the merging lane, respectively. The peaks of the graph show a high rate of energy dissipation indicating lane changing and overtaking maneuvers. Noticeable convergence of the variables at the final state implies the effectiveness of nonlinear controllers. With reference to Fig. 10, headings of a robot from the merging lane and a platoon of three robots can be seen facilitating lane merging, lane changing and overtaking maneuvers. The oscillatory nature of the graph of θ_2 shows the split maneuver of the follower robot, A_2 , while θ_4 shows the merge maneuver of the robot, A_4 .

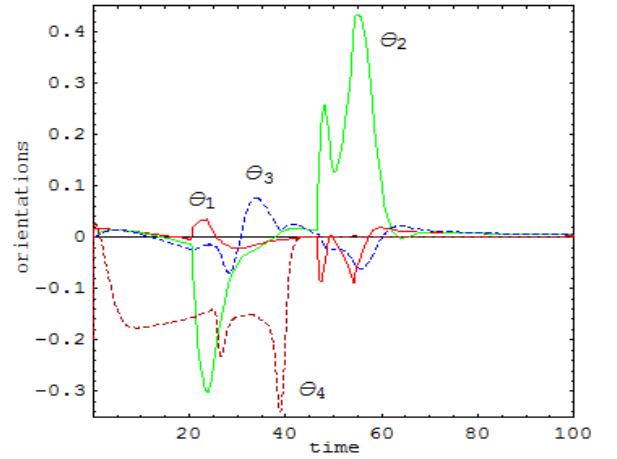
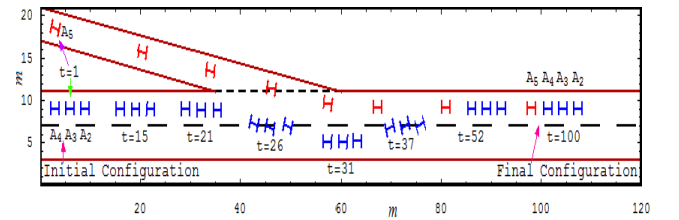
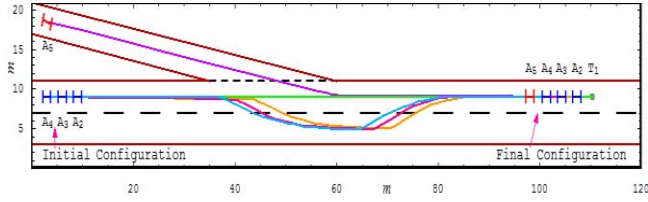


Fig. 10. The evolution of orientations of robots (θ_1 - leader robot and θ_4 - robot from merging lane).

B. Scenario 2: Platoon Overtaking via Virtual-Leader follower Scheme



(a) Platoon changes lane to give way to the mobile robot approaching from the merging lane. Robot from merging lane (A_5) is represented by red.



(b) shows the trajectory path generated by the platoon in the overtaking scenario. Path of the robot (A_5) from merging lane – represented in purple, T_1 is the target position of the virtual-leader and the path generated by this point (A_1) is represented in green.

Fig. 11 shows the simulation result of platoon overtaking.

Scenario 2 illustrates the implementation of a virtual-leader follower strategy to generate a lane change, merge and overtaking maneuvers by a 3-car platoon (see Fig. 11 (a)). The lane change maneuver is activated to give way to a mobile robot approaching from the merging lane. The entire platoon subsequently overtakes the mobile robot from the merging lane forming a new final platoon formation. The virtual-leader continues its path normally in the adjacent lane while a strategic selection of mobile ghost target for each follower robot allows the platoon to give way by changing its original lane and overtaking the robot looming from the merging lane. Fig. 11 (b) shows the path generated by the respective robots in the overtaking scenario.

The evolution and final convergence of nonlinear acceleration controllers of the virtual-leader is shown in Fig. 12. Once again, the acceleration components have similar convergence trends for the entire platoon.

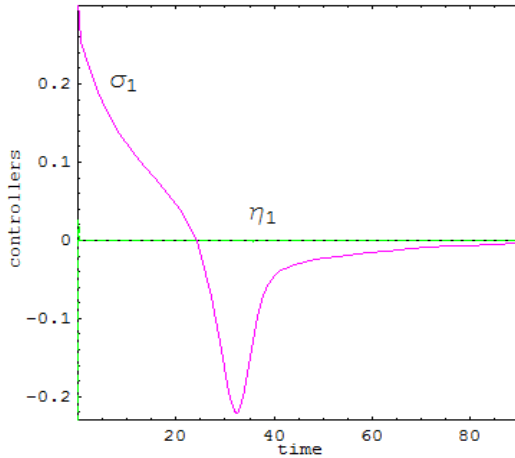


Fig. 12. Evolution of the acceleration components of A_1
(η_1 - translational, σ_1 - rotational).

XII. CONCLUDING REMARKS

A new artificial potential field method known as Lyapunov-based Control Scheme is utilized to solve the lane changing and overtaking maneuvers of a platoon of car-like robots. A set of continuous acceleration-based control laws incorporating a new leader follower strategy accomplish the desired tasks of this traffic problem. A virtual-leader is accommodated to activate and accomplish entire platoon lane change and overtaking maneuver. On the other hand, a real-leader follower strategy is adopted for a robot splitting from the platoon in the adjacent lane to give way to the mobile robot from the merging lane. Along the course of the journey, this robot overtakes the leader robot of the platoon and finally leads the new platoon formation. The numerical values (L_i and ψ_i) of the mobile ghost targets selected in piecemeal for specific stages achieve single robot and entire platoon lane changing (give-way) and overtaking maneuvers. It can be noted that the position of the leader robot is changed during the course of the journey, resulting a change in the leadership roles. The lane merging is accomplished by strategically assigning a separate target to the robot looming from the merging.

The new algorithm invariably helps nonholonomic car-like robots in a platoon formation to obtain safe and collision-free trajectories within a constrained environment. The controllers stabilize the configuration coordinates of the robot to an arbitrary small neighborhood of the target. The efficiency of the control laws have been demonstrated through some interesting simulations of lane changing and overtaking maneuvers. The successful application of the new control scheme can help form a theoretical basis of lane changing and overtaking problems of the IVS. The lane changing (give-way) and overtaking strategy at lane merge would reduce traffic congestion and possible road mishaps at the lane merge.

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